The Normal Distribution and the Central Limit Theorem

Unit 3: Why Averages Are Tame Even When the World Is Not

Stat 220 | Statistics for Data Science

Where this fits

- We can now describe a random quantity with a mean and a standard deviation.
- This part answers the question that makes statistics possible: **what does the distribution of an average look like?**
- The answer is the same bell curve almost every time, for reasons that have nothing to do with the data being "nice." That single fact powers every test, interval, and error bar you will ever report.

Principle: why the normal shows up

The bell curve is not common because nature loves it. It is common because *adding things up* produces it, and most of what we measure is a sum.

Live experiment: build a sampling distribution with your hands

Live demo: the class becomes the simulation

Everyone gets the same printed list of 200 house prices. It is badly right-skewed: mostly modest homes, a few mansions.

1. **Round 1:** close your eyes, point at **one** price. Write it down.
2. **Round 2:** pick **ten** prices at random and write down their *average*.

Come to the board and put a sticky note in the right bin: Round 1 values on the top axis, Round 2 averages on the bottom.

Materials: the printed price handout (`Unit_03_Price_Handout.pdf`), sticky notes, a board with two labeled axes. Time: 15 minutes. Before we look: predict which histogram will be more symmetric, and which will be wider.

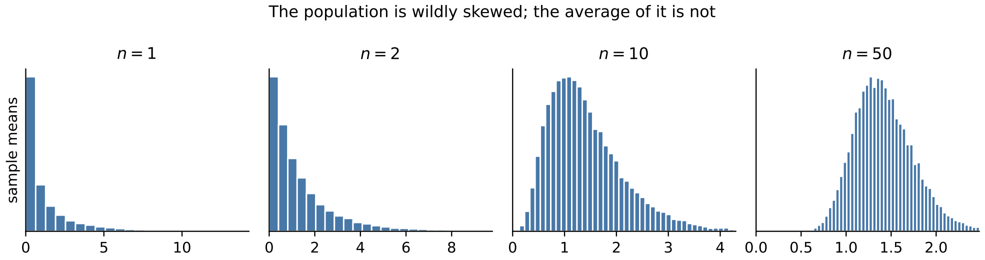
Stay here while they stick: what will happen

- The **top** histogram will look like the population: skewed, with a long right tail of mansions.
- The **bottom** histogram will be roughly symmetric, bell-shaped, and about $\sqrt{10} \approx 3.2$ times narrower.
- Nobody coordinated this. Averaging did it.

In practice: what the demo is really about

"Does my data have to be normally distributed to run a t-test?" The answer lives in the difference between those two histograms.

The reveal: the same thing, simulated 20,000 times



The population on the left is severely skewed. By $n = 10$ the average is already nearly symmetric, and by $n = 50$ it is a textbook bell. Notice also how the width collapses.

What this unit gives you

Things to know

- densities, and why $P(X = x) = 0$ for a continuous variable
- the normal distribution, z-scores, and 68–95–99.7
- the central limit theorem: what it promises and what it does not
- the standard error $\sigma/\sqrt{n}$, and why it drives every interval
- when the CLT is slow or useless: heavy tails, rare events, dependence

Mistakes to stop making

- testing the raw data for normality when the assumption is about the *estimate*
- quoting "$n > 30$" as if it were a law
- using 68–95–99.7 on skewed or heavy-tailed data
- confusing the SD of the data with the SE of an estimate
- counting rows when the observations are clustered

From bars to curves

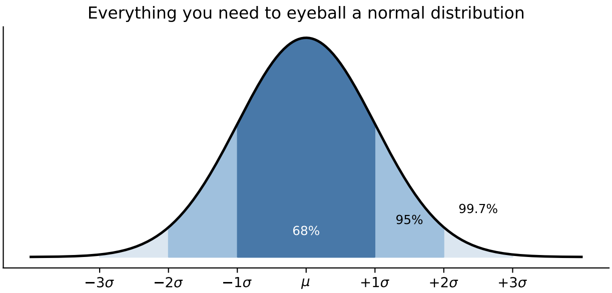
- For a continuous quantity (time, price, temperature) there are infinitely many possible values, so any exact value has probability zero.
- Instead we use a **density**: probability is the *area* under the curve between two points.
- "The chance a call lasts exactly 4.00000 minutes" is zero. "Between 3 and 5 minutes" is a real number, and it is an area.

Principle: ask for intervals, not points

Every meaningful question about a continuous variable is a question about a range. This is also why we report intervals rather than points when we estimate.

The normal distribution

Two parameters do all the work: the mean μ (where it sits) and the SD σ (how wide).



- About **68%** within 1σ.
- About **95%** within 2σ.
- About **99.7%** within 3σ.

Memorize these three numbers. They let you sanity-check any claim in your head.

How to read a density (it is not a probability)

- The height of the curve is **not** a probability. It can even exceed 1.
- Only *areas* are probabilities. The area under the whole curve is 1, and the area between two values is the chance of landing between them.
- Think of it as a histogram whose bars have been made infinitely thin: the bar heights stop being counts, but the area of a region still tells you what fraction lives there.

Principle: the practical consequence

Never ask "what is the probability that a call lasts exactly 4 minutes?" (it is zero). Ask for a range, or for a percentile. Every real question about a continuous quantity is one of those two.

The z-score: a universal ruler

$$z = \frac{x - \mu}{\sigma}$$ "how many standard deviations from the middle?"

- z has no units, so it compares things measured on different scales: a test score, a response time, a revenue figure.
- $|z| > 2$ is noteworthy, and $|z| > 3$ is rare, *if* the distribution really is normal.
- You have already met this: every t-statistic in this course is a z-score in disguise, "estimate minus null, divided by its standard error."

Principle: the signal-to-noise template returns

$\dfrac{\text{how big is the effect}}{\text{how much does it wobble}}$. That ratio is the whole of inference, and z is its simplest form.

Your turn #1

Your turn: read a z-score out loud

A logistics team says delivery times average 40 minutes with a standard deviation of 8 minutes. A customer's package took 70 minutes.

With a neighbor: compute the z-score, say how unusual it is if times were normal, and then give one reason the "if normal" clause might be doing a lot of work here.

Your turn #1: answer

- $z = (70 - 40)/8 = 3.75$. Under a normal model that is about a 1-in-10,000 delivery.
- But delivery times are **right-skewed**: traffic, weather, and failed handoffs create a long tail that a normal model does not have.
- So the honest reading is: unusual, but nowhere near one in ten thousand. The normal model is understating the tail.

Trap: sigma-counting on the wrong distribution

The 68–95–99.7 rule is a property of the normal curve, not of data in general. On skewed data it will call ordinary events miracles.

Three quick z-scores

Situation	value	z	read it as
Server response time, mean 220ms, SD 40ms	300ms	$+2.0$	slow, but a few percent of requests will be
Exam score, mean 74, SD 9	92	$+2.0$	same z, unrelated units
Daily revenue, mean $40k, SD $6k	$22k	-3.0	a genuinely bad day, *if* revenue is symmetric

Principle: why standardizing is worth the trouble

z strips the units off, so a response time and an exam score become comparable. Every test statistic in this course is a z-score with a different denominator.

The central limit theorem

If $X_1, \ldots, X_n$ are independent draws from *any* distribution with a finite mean μ and SD σ, then for large enough n:

$$\bar{x} \;\approx\; \text{Normal}\left(\mu,\; \frac{\sigma}{\sqrt{n}}\right).$$

- The shape of the original distribution does not appear anywhere in that statement.
- $\sigma/\sqrt{n}$ is the **standard error**: the typical distance between your sample mean and the truth.

Principle: this is why statistics works at all

We rarely know the distribution of what we measure. We do not need to. We only need the distribution of our *estimate*, and that one is predictable.

The distinction that everything hinges on

- **Not required:** that your data is normally distributed. Income, charges, and durations never are.
- **Required:** that your *estimate* (a mean, a difference, a coefficient) has an approximately normal sampling distribution.
- The CLT converts the second requirement into a condition about sample size and skew, not about the data's shape.

In practice: the sentence worth being able to say

"The t-test does not assume the data is normal. It assumes the sampling distribution of the mean is roughly normal, which the CLT delivers unless n is small or the data is extremely skewed."

The standard error, and the $\sqrt{n}$ tax

$$\mathrm{SD}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

- To halve your uncertainty you need **four times** the data. To cut it by ten, a hundred times.
- That is the arithmetic behind every "can we just collect more data?" conversation.
- It also explains why small samples produce such wild extremes, which is exactly the law of small numbers from the probability module.

Principle: precision is expensive and predictable

Before running a study, use $\sigma/\sqrt{n}$ to work out what precision your budget buys. That is the entire content of a power calculation.

Standard deviation versus standard error

Standard deviation (σ or s)

- spread of the **individual values**
- "how different are two random customers?"
- does *not* shrink with more data: a bigger sample estimates it better but does not make it smaller

Standard error ($\sigma/\sqrt{n}$)

- spread of your **estimate**
- "how far is my sample mean from the truth?"
- shrinks like $1/\sqrt{n}$

Trap: the error that produces absurd intervals

Using s where you needed $s/\sqrt{n}$ (or the reverse) is the single most common technical mistake students make all semester. Ask yourself out loud: *am I describing people, or describing my estimate?*

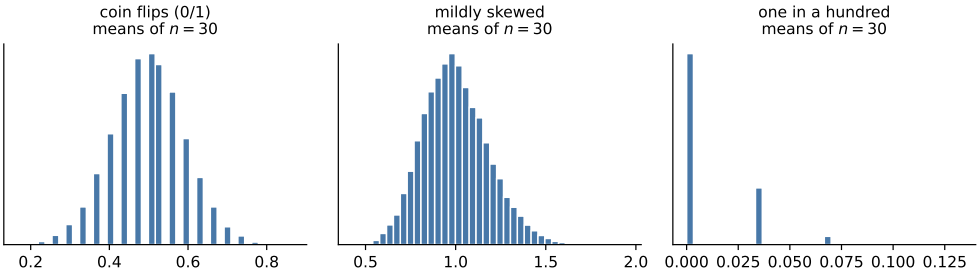
Four things the CLT does not say

- It does **not** say your data becomes normal. A bigger sample of incomes is still a pile of skewed incomes.
- It does **not** help with a *single* observation. Predicting one customer is governed by σ, not by $\sigma/\sqrt{n}$.
- It does **not** fix bias. If your sample is drawn from the wrong population, the average of it converges beautifully to the wrong number.
- It does **not** apply to every statistic. The sample maximum, for instance, never becomes normal no matter how much data you collect.

Principle: what it does say

Sums and averages of many independent, not-too-extreme pieces are approximately normal. That is narrow, and it is enough to build the whole course on.

How large is "large enough"? It depends on the skew



All three show means of $n = 30$. Coin flips are already perfect. Mild skew is fine. But a 1-in-100 rare event at $n = 30$ is still hopeless: most samples contain zero events, and the average is lumpy, not bell-shaped.

Retire the "$n > 30$" rule

- $n = 30$ is plenty for symmetric data, marginal for moderately skewed data, and useless for rare events or heavy tails.
- A workable rule for proportions: you want at least about 10 successes *and* 10 failures, not just 30 rows.
- If you are unsure, do not argue about rules: **bootstrap** the estimate and look at the shape of its distribution directly.

Trap: a rule of thumb quoted as a law

"$n > 30$ so the CLT applies" is the most confidently repeated wrong sentence in applied statistics. The right question is always: how skewed, and how rare?

Your turn #2

Your turn: is the CLT going to save you?

Three analyses, each with $n = 40$:

1. average height of 40 students
2. average purchase amount of 40 customers, where one bought a $45,000 tractor
3. fraction of 40 users who clicked, when the click rate is about 0.5%.

With a neighbor: for each, will the sample mean be roughly normal? Rank them from safest to most dangerous.

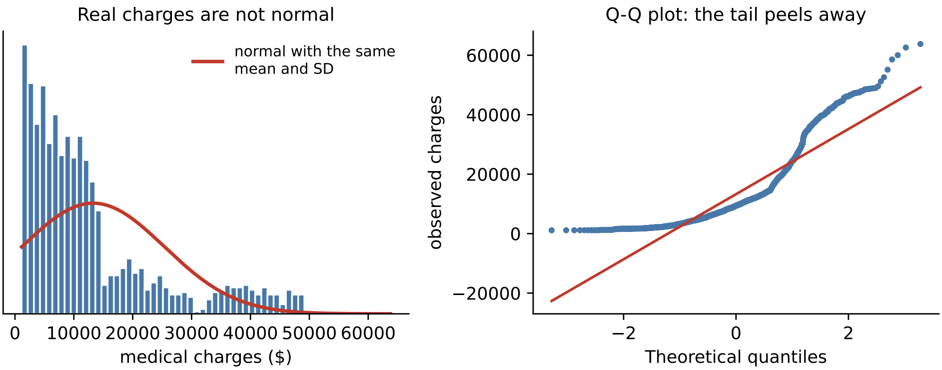
Your turn #2: answer

- **Heights** (safest): nearly symmetric already, so $n = 40$ is more than enough.
- **Purchases**: one enormous value dominates the mean. The sampling distribution is skewed and the SE is unstable. Report a median, or bootstrap, or model the tail.
- **Clicks** (most dangerous): with 0.5% and $n = 40$ you expect *0.2* clicks. There is no meaningful normal approximation. You need thousands of users, or an exact binomial method.

Principle: the CLT is about the effective number of informative observations

Forty rows containing one event is not a sample of forty.

Real data is not normal, and that is fine



Medical charges for 1,338 people. The normal curve with the same mean and SD predicts negative charges and misses the entire right tail. The Q-Q plot shows the tail peeling away from the line. None of this stops us from estimating the *mean* charge with a normal-based interval.

Reading a Q-Q plot in five seconds

- Points on the line: consistent with normal.
- Both ends curving **up** above the line on the right: a long right tail (money, durations, city sizes).
- An S-shape: heavy tails on both sides, so extremes are more common than normal predicts.
- Steps or gaps: the variable is discrete or has been rounded or censored.

Principle: diagnose, do not just test

A normality *test* gives you a p-value that is useless at both extremes: it rejects harmless deviations in huge samples and misses real ones in small samples. Look at the picture instead.

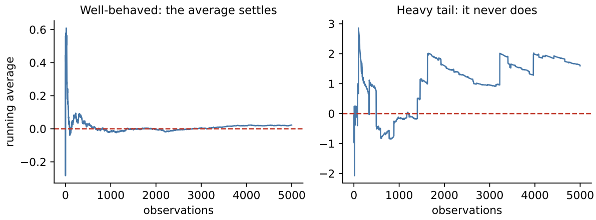
The log-normal: when things multiply

- The CLT says sums of many small effects become normal. If effects *multiply* instead of adding, the logarithm becomes normal and the variable is **log-normal**.
- Income, revenue per customer, insurance claims, city sizes, and stock prices are all roughly log-normal, because growth compounds.
- That is why taking logs so often fixes a skewed regression: it converts a multiplicative world into an additive one.

In practice: a useful way to say it

"Money is usually log-normal, not normal, because it grows by percentages rather than by increments."

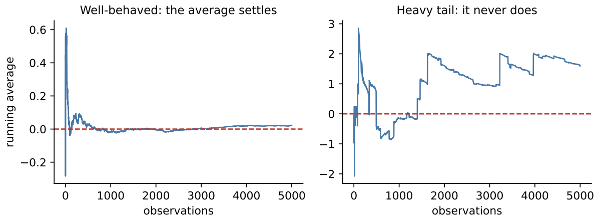
Heavy tails: where the argument breaks entirely



On the left, a well-behaved variable: the running average locks onto the truth.

On the right, a heavy-tailed one: every so often a single enormous draw resets the average. It never converges, because the theoretical mean does not exist.

Heavy tails: where the argument breaks entirely



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Insurance losses, market crashes, and viral traffic live closer to the right panel than anyone likes.

Dependence quietly destroys $\sqrt{n}$

- The CLT assumes independent draws. Real data is full of dependence: repeated measurements on the same user, days in a row, students in the same class.
- With dependence, your **effective** sample size is far below your row count, so the standard error you computed is too small and your interval is too narrow.
- Symptom: absurdly tiny p-values on data that everyone knows is clustered. You saw this in Part 1, with 400,000 sessions from two cities.

Trap: counting rows instead of independent units

Ask "how many independent things do I really have?" Often the answer is the number of users, or stores, or weeks, not the number of rows.

Your turn #3

Your turn: which number does the question need?

Your company's delivery times have mean 42 minutes and SD 15 minutes, measured on 900 deliveries. For each question, say whether you need the **SD** or the **SE**, and compute it:

1. "Roughly what range covers most individual deliveries?"
2. "How precisely do we know the average delivery time?"
3. "Should we promise every customer delivery within an hour?"

Your turn #3: answer

- (1) **SD.** About 95% of individual deliveries fall within roughly $42 \pm 2(15)$, so 12 to 72 minutes. Wide, because people vary.
- (2) **SE.** $15/\sqrt{900} = 0.5$ minutes, so the average is known to about 42 ± 1 minute. Narrow, because averaging kills noise.
- (3) **SD again**, and it is bad news: an hour is only $z = 1.2$ above the mean, so roughly 12% of deliveries would miss the promise. Knowing the average within a minute does not help a single late customer.

Principle: the question to ask yourself

Am I describing *people*, or describing *my estimate*? Promises to customers need the SD. Claims about a population average need the SE.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM about an assumption

We told it: "Our revenue-per-customer data is very right-skewed. We have 5,000 customers. Can we use a t-test to compare two groups?"

It answered: "No. The t-test requires normally distributed data. Since your data is skewed, you must transform it or use a nonparametric test instead."

Mistake, or no mistake?

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Mistake, or no mistake?

Trap: verdict: mistake

The assumption is about the sampling distribution of the *mean*, not the raw data. With 5,000 customers per group the CLT handles ordinary skew comfortably. (A transformation might still be a good idea, but for interpretability or heavy tails, not because the test is invalid.)

LLM check: mistake or not? (2 of 2)

LLM check: same territory, a different answer

We told it: "We ran 40 users and 0.5% typically click. Our A/B test shows a 50% lift. Is it significant?"

It answered: "With a 0.5% base rate and 40 users you expect a fraction of one click, so the normal approximation does not apply and almost no result would be interpretable. I would use an exact binomial method, and more importantly, I would compute the sample size needed to detect a 50% lift, which will be in the tens of thousands."

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Mistake, or no mistake?

Principle: verdict: no mistake

It recognized that $n = 40$ is not $n = 40$ when the event is rare, refused the approximation, and redirected to the real question: how much data would this take?

Deciding whether your normal-based answer is safe

1. **Name the estimate**, not the data: is it a mean, a difference, a proportion, a coefficient?
2. **Plot the raw data** once, to see skew, gaps, and outliers. Do not test it.
3. **Count the informative observations**: successes and failures, not just rows.
4. **Ask what is independent**, and reduce n to that.
5. **If in doubt, bootstrap** the estimate and look at the shape you actually get.
6. **Report the SE and the interval**, and say which assumption is carrying the weight.

Scenario: the outlier that ate the interval

Setup. An analyst reports average order value of \$310 with a 95% interval of \$180 to \$440 from 200 orders. One order was a \$40,000 bulk purchase.

- That single order dominates both the mean and the SD, so the interval is driven by one observation rather than by 200.
- Deleting it silently is not an option: bulk orders are real revenue.
- Better: report the median alongside the mean, bootstrap the interval, and, if the business question is about typical customers, analyze bulk orders as their own segment.

In practice: how to answer this

"I would ask whether the question is about the total (keep the whale) or the typical customer (model it separately), and I would use a bootstrap rather than the formula."

Deciding whether a normal-based answer is safe

1. **Name the estimate**, not the data: a mean, a difference, a proportion?
2. **Plot the raw data once** to see skew, gaps, and outliers. Do not run a normality test.
3. **Count the informative observations**, which for a proportion means successes and failures, not rows.
4. **Ask what one independent observation is**, and reduce n to that.
5. **Convert to a z-score** when you need to compare across units: $z = (x - \mu)/\sigma$.
6. **If in doubt, simulate.** Resample your own data and look at the shape of the estimate.
7. **Report the SE and the interval**, and say which assumption is carrying the weight.

The traps, all in one place

Trap: the ones that bite most often

- Requiring the raw data to be normal when the assumption is about the estimate.
- Reciting "$n > 30$" regardless of skew or rarity.
- Applying the 68–95–99.7 rule to skewed or heavy-tailed data.
- Treating 3-sigma events as impossible in fat-tailed domains.
- Using row counts as n when the observations are clustered or repeated.
- Running a normality test and letting its p-value make the decision.

Ten questions to be able to answer

1. Why is the normal distribution so common?
2. Does a t-test require normally distributed data? Explain carefully.
3. What is a standard error, and how does it differ from a standard deviation?
4. How large does n need to be for the CLT to apply?
5. What does a z-score of 3 mean, and when would you not be impressed by one?
6. Our data is log-normal. What does that suggest about how it was generated?
7. Why do heavy tails break the usual averaging argument?
8. We have 100,000 rows but only 300 customers. What is our real sample size?
9. How would you check whether a normal-based interval is trustworthy here?
10. A colleague deletes outliers until a normality test passes. What do you say?

Mastery checklist

You have this unit when you can, from memory:

- explain density, area as probability, and why point probabilities are zero
- compute and interpret a z-score, and use 68–95–99.7 to sanity-check claims
- state the CLT precisely, including what it assumes
- explain why the standard error is $\sigma/\sqrt{n}$ and what that costs you
- distinguish normality of the data from normality of the estimate
- name three situations where the CLT will not rescue you

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Normal probabilities and percentiles	`stats.norm.cdf`, `stats.norm.ppf`
Standardize a value	`(x - mean) / sd`
Check normality by eye	`stats.probplot`, plus a histogram
Build a sampling distribution yourself	`rng.integers` to resample, then `.mean(axis=1)`
Standard error of a mean	`s / np.sqrt(n)`
Check whether an interval really covers 95%	simulate many samples and count
Measure dependence in a series	`Series.autocorr(1)`

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- We now know the shape of an estimate's uncertainty. Next: **estimation itself**, how to build a good estimate, judge it, and put an interval around it without leaning on a formula.
- That module introduces likelihood and the bootstrap, and it is the direct on-ramp to the inference material.

Principle: the through-line

You almost never know the distribution of what you measure. You can almost always know the distribution of what you estimate. Statistics lives in that gap.